



Lab 07

SE3082 - Parallel Computing

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Timing Results

Exercise	Program	2 processes	4 processes	8 processes
Exercise 1	Broadcast + Send Recv	0.002674 sec	0.005507 sec	0.005855 sec
Exercise 2	Scatter + Send Recv	0.001699 sec	0.001895 sec	0.000936 sec
Exercise 3	Scatter + Gather	0.001418 sec	0.001047 sec	0.003952 sec
Exercise 4	Scatter + Reduce	0.002761 sec	0.002522 sec	0.001448 sec
Exercise 5	Scatter + Allreduce	0.001558 sec	0.003906 sec	0.002644 sec
Exercise 6	Scatter + Scan	0.001406 sec	0.001559 sec	0.000805 sec

All measurements are elapsed times reported by MPI_Wtime. Small timing differences between runs are normal because process startup, operating system scheduling, and the local machine load affect short programs.

Collective Communication Comparison

Program	Collectives	Array storage	Manual sum	Result available
sum_bcast	MPI_Bcast, MPI_Send, MPI_Recv	Full array on every process	Yes	Root only
sum_scatter	MPI_Scatter, MPI_Send, MPI_Recv	Full array on root; chunk per process	Yes	Root only
sum_gather	MPI_Scatter, MPI_Gather	Full array on root; chunk per process	Yes	Root only
sum_reduce	MPI_Scatter, MPI_Reduce	Full array on root; chunk per process	No	Root only
sum_allreduce	MPI_Scatter, MPI_Allreduce	Full array on root; chunk per process	No	Every process
sum_scan	MPI_Scatter, MPI_Scan	Full array on root; chunk per process	No	Prefix result per rank

Answers to Exercise 7

As scatter does not involve sending all the array to each process, it consumes less memory and sends less information compared to broadcast. While gather makes it easier to receive partial sums, root still has to sum them sequentially. In reduce, the operation is collective, and this approach is suitable when only the root needs the result. Allreduce is used when each process needs the same global sum.

The quickest execution time for scatter + scan with 8 processes was 0.000805 sec. However, these programs work for only several milliseconds; thus, each run is affected by system noise. In general, the tendency can be stated as scatter being more efficient than broadcast in terms of memory, and reduction being faster due to lack of summation loop.

MPI_Scan is used rather than MPI_Allreduce when each rank needs a different cumulative sum. For instance, each rank writes a varying number of records to one output array, and the prefix sum provides each rank with an initial offset. MPI_Allreduce cannot provide an offset because each rank gets the same global sum.

Verification

Each program calculated the expected sum of the first 1,000,000 integers, which was 500000500000. Exercise 5 provided the sum on every rank. Exercise 6 confirmed the correctness of the rank prefix sum and displayed the correct total at the last rank.